

HOMework 1

ELEMENTARY LINEAR ALGEBRA

(MATH 2250-03), FALL, 2018

CHECK: 02/08/2019 DUE: 02/11/2019

SECTION 1.1 - SYSTEMS OF LINEAR EQUATIONS

1. Find the augmented matrix associated to the following system of linear equations. Then, using elementary row operations, solve this system.

$$2x_1 + 4x_2 = -4$$

$$5x_1 + 7x_2 = 11$$

2. Determine if the following system of linear equations is consistent by finding the augmented matrix associated to the system and solving the system using elementary row operations.

$$x_1 + 3x_3 = 2$$

$$2x_2 - 3x_4 = 3$$

$$-2x_2 + 3x_3 + 2x_4 = 1$$

$$3x_1 + 7x_4 = -5$$

3. Do the three planes given by the linear equations $x_1 + 2x_2 + x_3 = 4$, $x_2 - x_3 = 1$ and $x_1 + 3x_2 = 0$ have at least one common point of intersection?
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SECTION 1.2 - ROW REDUCTION AND ECHELON FORMS

4. Using elementary row operations, compute the reduced row echelon form of the following matrix:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix}$$

5. Find the general solution of the system of linear equations that corresponds to the following augmented matrix:

$$\begin{bmatrix} 1 & -3 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

6. Determine the value(s) of $h \in \mathbb{R}$ such that the following matrix is the augmented matrix of a consistent linear system:

$$\begin{bmatrix} 2 & 3 & h \\ 4 & 6 & 7 \end{bmatrix}$$

SECTION 1.3 - VECTOR EQUATIONS

7. Let $a_1 = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$, $a_2 = \begin{bmatrix} -2 \\ -3 \\ 7 \end{bmatrix}$, and $b = \begin{bmatrix} 4 \\ 1 \\ h \end{bmatrix}$. For what value(s) of $h \in \mathbb{R}$ is $b \in \text{span}\{a_1, a_2\}$?

8. Let $v_1, \dots, v_k \in \mathbb{R}^3$ and suppose that for $j = 1, \dots, k$ an object with mass $m_j \in \mathbb{R}$ is located at v_j . Physicists call such objects *point masses*. The total mass of this system of point masses is $m = m_1 + \dots + m_k$, and the center of mass of the system is $\bar{v} = \frac{1}{m}(m_1 v_1 + \dots + m_k v_k)$. Find the center of mass for the following system of point masses:

$$v_1 = \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 4 \\ 3 \\ -2 \end{bmatrix}, \quad v_3 = \begin{bmatrix} -4 \\ -3 \\ -1 \end{bmatrix}, \quad v_4 = \begin{bmatrix} -9 \\ -8 \\ 6 \end{bmatrix}$$

$$m_1 = 2g, \quad m_2 = 5g, \quad m_3 = 2g, \quad m_4 = 1g$$